

Neha Yadav

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Summary

- Software Developer with 5+ years of experience architecting large-scale, enterprise-grade AI, cloud, and microservices systems across healthcare, finance, defense, and education, delivering measurable operational impact and 50%+ efficiency gains.
- Expert in AI/ML, microservices, event-driven architectures, and cloud-native systems, with hands-on experience in TensorFlow, Python, Java, Kafka, Redis, and AWS/Azure, powering predictive analytics, fraud detection, and mission-critical enterprise platforms.
- Designed and deployed secure, high-performance distributed systems with JWT/OAuth2 authentication, AES-256 encryption, and multi-region cloud deployments, achieving zero security breaches and 99.99% uptime.
- Built automated CI/CD and DevOps pipelines, integrating Terraform, Docker, Kubernetes, Jenkins, Prometheus, and Grafana, compressing deployment cycles by 30–40% and enhancing system reliability under high concurrency.
- Delivered real-time, high-throughput data pipelines and predictive analytics platforms, processing millions of events/hour, enabling data-driven decision-making, operational scaling, and measurable business outcomes across diverse enterprise domains.

Technical Skills

Languages: Java, Python, C#, C++, JavaScript/TypeScript, Go, Kotlin, Bash, PowerShell

Core CS: Data Structures & Algorithms, OOP, Design Patterns, System Design, Concurrency

Architecture: Microservices, Event-Driven Systems, REST/GraphQL APIs, Scalable & Fault-Tolerant Design

Databases: MySQL, PostgreSQL, MongoDB, DynamoDB, Redis, Cassandra; Query Optimization & Replication

Cloud & DevOps: AWS, Azure, GCP, Docker, Kubernetes, Terraform, CI/CD (Jenkins, GitHub Actions), Git

Testing: Unit & Integration Testing (JUnit, PyTest, NUnit), API/UI Testing (Postman, Selenium), Code Quality

Web: HTML5, CSS3, React, Angular, Vue.js, Responsive Design, UI/UX Principles

Security & Performance: OAuth2, JWT, Encryption, Caching, Performance Optimization

Methodologies: Agile (Scrum/Kanban), Cross-Functional Collaboration, Documentation

Experience

Software Developer | Centric Solutions, USA

Jul 2023 – Present

- Engineered AI-driven recruitment and predictive maintenance platforms leveraging Java, Python, and TensorFlow, increasing predictive accuracy by 35% and accelerating operational decision-making across 50k+ concurrent users.
- Architected distributed microservices ecosystem with Spring Boot, Kafka, and Redis, reducing system latency by 40% while supporting real-time analytics for flight operations and fraud detection.
- Designed and deployed secure EHR and battlefield asset tracking solutions with AES-256 encryption, JWT/OAuth2 authentication, and AWS IAM policies, achieving zero security breaches in 18 months.
- Automated CI/CD pipelines and infrastructure provisioning using Jenkins, GitHub Actions, Terraform, and Docker/Kubernetes, cutting deployment time by 30% and enabling multi-environment scalability.
- Implemented real-time data processing and streaming pipelines with Kafka and AWS Lambda, enhancing fraud detection and predictive analytics throughput by 50%.
- Optimized backend database architectures (MySQL, PostgreSQL, MongoDB) with advanced indexing, sharding, and query tuning, improving transactional performance by 45% under high concurrency.
- Developed and integrated RESTful and GraphQL APIs with robust unit and integration testing using JUnit and PyTest, improving service reliability and reducing bug reports by 25%.

Software Developer | Hexaware Technologies, India

Jul 2019 – Nov 2021

- Built AI-powered loan eligibility and claims automation engines using Python, Java, and ML pipelines, reducing manual review by 60% and improving approval speed by 50%, impacting millions in processed transactions.
- Architected cloud-native, containerized platforms on AWS & Azure with Docker, Kubernetes, and Terraform, enabling auto-scaling to 100k+ users with 99.99% uptime.
- Designed high-throughput, event-driven microservices architectures for portfolio management and e-learning platforms, leveraging Kafka and Redis Streams to process 1M+ events/hour with 100ms latency.
- Spearheaded end-to-end DevOps and observability strategy, integrating CI/CD, GitOps, Prometheus, Grafana, and CloudWatch, reducing incident response time by 40%.
- Delivered secure, high-performance data pipelines integrating SQL/NoSQL databases, implementing advanced caching, indexing, and ACID-compliant transaction handling, improving query efficiency by 50%.
- Led front-end innovation with React, TypeScript, and responsive design, creating personalized e-learning and clinical trial dashboards that boosted user engagement by 30% and reduced bounce rates.
- Collaborated with cross-functional teams to implement automated reporting and analytics dashboards using Power BI and Tableau, enabling data-driven decision-making and reducing reporting cycle time by 35%.

Education

Master of Science in Information Systems

Jan 2022 – May 2023

Cleveland State University, USA

Bachelor of Technology in Computer Science

Aug 2016 – Aug 2020

Aurora Engineering College, India